

CLASS 9 · CBSE · GANITA MANJARI PART 1 · DPP SHEET

Chapter 6: Measuring Space: Perimeter and Area

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the circumference of a circle of radius 10.5 cm. Take $\pi = \frac{22}{7}$.
2. The circumference of a circle is 132 cm. Find its radius. Take $\pi = \frac{22}{7}$.
3. Find the length of an arc of radius 14 cm that subtends 45° at the centre. Take $\pi = \frac{22}{7}$.
4. Find the perimeter of a semicircular disc of radius 7 cm. Take $\pi = \frac{22}{7}$.
5. Use Heron's formula to find the area of a triangle with sides 5 cm, 12 cm and 13 cm.
6. Find the area of a triangle with sides 10 cm, 17 cm and 21 cm.
7. Find the area of an equilateral triangle of side 6 cm.
8. The diagonals of a rhombus are 16 cm and 30 cm. Find its area and its side.
9. Find the area of a sector of radius 14 cm and angle 45° . Take $\pi = \frac{22}{7}$.
10. A chord of a circle of radius 7 cm subtends 90° at the centre. Find the area of the minor segment. Take $\pi = \frac{22}{7}$.

Answer Key

1. 66 cm
2. 21 cm
3. $\frac{45}{360} \times 88 = 11$ cm
4. $22 + 14 = 36$ cm
5. $s = 15$; $\sqrt{15 \times 10 \times 3 \times 2} = 30$ cm²
6. $s = 24$; $\sqrt{24 \times 14 \times 7 \times 3} = 84$ cm²
7. $9\sqrt{3}$ cm²
8. Area 240 cm²; side $\sqrt{8^2 + 15^2} = 17$ cm
9. $\frac{1}{8} \times 616 = 77$ cm²
10. $38.5 - 24.5 = 14$ cm²